

Ran Li, Ph.D.

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Research Interests

Computational medical imaging; quantitative MRI; deep learning for physiological parameter estimation; physics-informed machine learning; multimodal image analysis (MR/PET/CT/Ultrasound); medical image segmentation and reconstruction; real-time surgical image guidance; AI-assisted clinical decision systems; GPU-accelerated deployment.

Education

Washington University in St. Louis, St. Louis, MO

Ph.D., Imaging Science

Sep 2019 – May 2024

GPA: 3.9

Washington University in St. Louis, St. Louis, MO

M.S., Electrical Engineering

Sep 2015 – May 2017

Xi'an University of Technology, Xi'an, China

B.S., Electrical Engineering

Sep 2010 – Jul 2014

Academic & Professional Appointments

EDDA Technology, Princeton, NJ

Research Scientist

Feb 2025 – Present

- Develop AI-driven surgical planning and intraoperative image guidance systems integrating pre-operative imaging with real-time endoscopic video.
- Lead deployment and optimization of GPU-accelerated inference and visualization pipelines on NVIDIA edge platforms (IGX Orin).

Biomedical MR Center, Washington University in St. Louis, St. Louis, MO

Postdoctoral Research Associate

May 2024 – Jan 2025

- Advanced deep learning and quantitative imaging pipelines for MRI/PET studies focused on perfusion, metabolism, and vascular disease.

Biomedical MR Center, Washington University in St. Louis, St. Louis, MO

Ph.D. Researcher

Sep 2019 – May 2024

- Conducted research in quantitative MRI and deep learning under faculty mentorship; developed models validated with in vivo standards.

Biomedical MR Center, Washington University in St. Louis, St. Louis, MO

Research Assistant

Sep 2017 – May 2019

- Developed customized software packages for MR image analysis; built toolkits for denoising, registration, and 3D reconstruction.

Research Experience (Selected Projects)

DeepMASL: Deep learning-enabled myocardial blood flow mapping (MRI ASL) 2023 – 2025

- Designed regression networks for myocardial blood flow (MBF) mapping from arterial spin labeling (ASL) cardiac MRI.
- Addressed limited ground truth using physics-informed simulation and augmentation; improved robustness to acquisition variability.
- Validated against microsphere-derived perfusion measurements in vivo; contributed to end-to-end quantitative pipeline design.

Deep learning quantification of myocardial oxygen extraction fraction (OEF) on non-contrast MRI 2021 – 2024

- Developed deep learning methods to estimate myocardial OEF and related metabolic metrics from noncontrast MRI.
- Built analysis workflows for reproducibility evaluation across subjects and acquisition settings; supported translational studies.

Two-stage carotid plaque segmentation with histology-aligned training 2021 – 2023

- Developed coarse-to-fine neural network pipeline for carotid plaque segmentation using multi-weighted MRI.
- Improved component-level segmentation under class imbalance and limited labels; enabled higher-resolution training supervision.
- Resulted in US provisional patent filing and peer-reviewed publication.

Quantitative multi-modal imaging pipelines (MR/PET/Ultrasound) 2018 – 2025

- Implemented automated pipelines for perfusion mapping, OEF estimation, inflammation analysis, and ultrasound elastography studies.
- Built toolchains using Python/MATLAB/ITK/ANTs for preprocessing, registration, segmentation, and statistical reporting.

AR-assisted endoscopic surgical guidance and edge deployment 2025 – Present

- Designed AR overlay pipeline integrating preoperative segmentation and intraoperative 4K endoscopic video for surgical guidance.
- Deployed C++/LibTorch inference pipelines with CUDA acceleration on NVIDIA IGX Orin; performed profiling and latency optimization.
- Built containerized runtime with Docker for reproducible deployment and system integration in safety-critical environments.

Peer-Reviewed Journal Publications

1. Nie X, Elvington A, Laforest R, Zheng J, Voller TF, Zayed MA, Abendschein DR, Bandara N, Xu J, **Li R**, Randolph GJ, Gropler RJ, Lapi SE, Woodard PK. ⁶⁴Cu-ATSM Positron Emission

- Tomography/Magnetic Resonance Imaging of Hypoxia in Human Atherosclerosis. *Circulation: Cardiovascular Imaging*. 2020;13(1):e009791.
2. Lu L, Eldeniz C, An H, **Li R**, Yang Y, Schindler TH, Peterson LR, Woodard PK, Zheng J. Quantification of myocardial oxygen extraction fraction: A proof-of-concept study. *Magnetic Resonance in Medicine*. 2021;85:3318–3325.
 3. Zheng J, **Li R**, Zayed MA, Yan Y, An H, Hastings MK. Pilot study of contrast free MRI reveals significantly impaired calf skeletal muscle perfusion in diabetes with incompressible peripheral arteries. *Vascular Medicine*. 2021;26:367–373.
 4. Zheng J, Sorensen C, **Li R**, An H, Hildebolt CF, Zayed MA, Mueller MJ, Hastings MK. Deteriorated regional calf microcirculation measured by contrast-free MRI in patients with diabetes mellitus and relation with physical activity. *Diabetes & Vascular Disease Research*. 2021;18:14791641211029002.
 5. Wang K, Zhang W, Li S, Jin H, Jin Y, Wang L, **Li R**, Yang Y, Zheng J, Cheng J. Noncontrast T1 dispersion imaging is sensitive to diffuse fibrosis: A cardiovascular magnetic resonance study at 3T in hypertrophic cardiomyopathy. *Magnetic Resonance Imaging*. 2022;91:1–8.
 6. Zheng J, **Li R**, Dickey EE, Yan Y, Zayed MA, Zellers JA, Hastings MK. Regional skeletal muscle perfusion distribution in diabetic feet may differentiate short-term healed foot ulcers from non-healed ulcers. *European Radiology*. 2023. doi:10.1007/s00330-023-09405-6.
 7. **Li R**, Edalati M, Muccigrosso D, Lau JMC, Laforest R, Woodard PK, Zheng J. A Simplified Method to Correct Saturation of Arterial Input Function for Cardiac Magnetic Resonance First-Pass Perfusion Imaging: Validation with simultaneously acquired PET. *Journal of Cardiovascular Magnetic Resonance*. 2023;25(1):35.
 8. **Li R**, Zheng J, Saffitz JE, Woodard PK, Jha AK. Carotid atherosclerotic plaque segmentation in multi-weighted MRI using a two-stage neural network: Advantages of training with high-resolution imaging and histology. *Frontiers in Cardiovascular Medicine*. 2023.
 9. Wahidi R, Zhang Y, **Li R**, Zayed MA, Hastings MK, Zheng J. Quantitative Assessment of Peripheral Oxidative Metabolism with a new Dynamic 1H MRI: a Pilot Study in People with and without Diabetes Mellitus. *Journal of Magnetic Resonance Imaging*. 2023.
 10. Zhang XL, Heo GS, Li A, Lahad D, Detering L, Tao J, Gao XF, Zhang XH, Luehmann H, Sultan D, Lou L, Venkatesan R, **Li R**, Zheng J, Amrute J, Lin CY, Kopecky BJ, Gropler RJ, Bredemeyer A, Lavine K, Liu YJ. Development of a CD163 Targeted PET Radiotracer Imaging Resident Macrophages in Atherosclerosis. *Journal of Nuclear Medicine*. 2024. (Accepted.)
 11. Zellers J, **Li R**, Vaidya R, Lohse K, North A, Cui S, Houston B, Chen M, Zheng J, Baxter J. Effect of scanning parameters on ultrasound shear wave elastography variability in tendon. *Knee Surgery, Sports Traumatology, Arthroscopy*. 2024.
 12. **Li R**, Eldeniz C, Wang K, Nguyen N, Schindler TH, Huang Q, Peterson LR, Yang Y, Yan Y, Cheng J, Woodard PK. Quantification of myocardial oxygen extraction fraction on noncontrast MRI enabled by deep learning. *Radiology Advances*. 2024 Nov;1(4):umae026.
 13. Huang Q, Tang H, Wang K, **Li R**, Eldeniz C, Nguyen N, Schindler TH, Peterson LR, Yang Y, Cheng J. Model-based self-supervised learning for quantitative assessment of myocardial oxygen extraction fraction and myocardial blood volume. *Magnetic Resonance in Medicine*. 2025 May 5.

14. Dai L, Wang K, **Li R**, Qiao J, Berberet C, Huang Q, Ji J, Bugenhagen SM, Schindler TH, Peterson LR, Woodard PK. Altered Myocardial Oxygenation and Impaired Energy Efficiency: A Pilot Study in Patients With Light Chain Cardiac Amyloidosis. *Journal of Magnetic Resonance Imaging*. 2025 Oct 14.
15. **Li R**, Berberet C, Huang Q, Woodard PK, Zheng J. Improved accuracy for myocardial blood flow mapping with deep learning-enabled CMR arterial spin labeling (DeepMASL): validation by microsphere in vivo. *Journal of Cardiovascular Magnetic Resonance*. 2025 Nov 14:101989.

Conference Abstracts & Presentations

1. Edalati MA, Hastings MK, Mohamed ZA, Muccigrosso D, **Li R**, Mueller MJ, Zheng J. MRI Methods for Exercise-based Perfusion Assessment of Diabetic Feet with Ulcers. *ISMRM*, 2018, Paris, France.
2. Zheng J, Gharabaghi S, **Li R**, Chen YS, An H, Haacke EM, Zayed MA, Hastings MK. Pilot contrast free MRI reveals significantly impaired calf skeletal muscle perfusion in diabetes. *ISMRM*, 2021 (Virtual).
3. Wahidi R, Zhang Y, **Li R**, Zayed MA, Hastings MK, Zheng J. Quantitative assessment of peripheral metabolism and perfusion with dynamic 1H MRI. *ISMRM*, 2022, London, UK.
4. **Li R**, Zheng J, Woodard PK, Jha AK. Carotid plaque segmentation in multi-weighted MRI using a two-stage neural network. *ISMRM*, 2022, London, UK.
5. Wang K, Zhang WB, Li S, Jin HR, Jin Y, Wang L, **Li R**, Yang Y, Cheng JL, Zheng J, An J. T1 dispersion imaging detects diffuse fibrosis in HCM. *ISMRM*, 2022, London, UK.
6. Mei BC, **Li R**, Eldeniz C, Schindler T, Peterson LR, Woodard PK, Zheng J. Deep Learning Assisted CMR Myocardial OEF Measurements. *SCMR*, 2023, San Diego, CA.
7. Wang J, **Li R**, Fu X, Yang Y, Sun XJ, Shu S, Zhao J, Kong XC, Zheng J. OEF correlates with fibrosis after heart transplant. *ISMRM*, 2023, Toronto, Canada.
8. **Li R**, Eldeniz C, Schindler TH, Peterson LR, Woodard PK, Zheng J. Deep learning assisted quantification of myocardial OEF. *ISMRM*, 2023, Toronto, Canada. (Travel Award)
9. **Li R**, Kaminski O, Eldeniz C, Schindler T, Peterson L, Woodard P, Zheng J. AI-based quantification of myocardial OEF and blood volume: reproducibility and distribution. *SCMR*, 2024, London, UK. (Oral/In-person)
10. **Li R**, Zheng J. Quantification of myocardial blood flow with deep learning-assisted cardiac ASL: validation by microsphere. *ISMRM*, 2024, Singapore.
11. **Li R**, Eldeniz C, Schindler T, Peterson L, Woodard P, Zheng J. AI-based quantification of myocardial OEF and blood volume: reproducibility, sex, and homogeneity. *ISMRM*, 2024, Singapore.

Patents

- “Carotid Plaque Segmentation Using a Trained Neural Network,” US Provisional Application 63/399,325, filed Aug 19, 2022 (**Li R.**, Jha A., Zheng J., Woodard P.)

Awards & Honors

- ISMRM Travel Award (for in-person presentation), 2023.

Teaching Experience

Washington University in St. Louis

Teaching Assistant, Mathematics in Imaging Science

Jul 2021 – Dec 2021

- Conducted 80+ hours of individual and group tutoring sessions using presentations and code demonstrations.
- Graded assignments and exams covering linear vector spaces, operator theory on Hilbert spaces, and applied functional analysis.

Professional Service

- Peer reviewer for scientific journals (medical imaging / MRI / ML).

Technical Skills

- **Programming:** Python, C++, CUDA, MATLAB, Bash
- **ML & CV:** Deep learning, segmentation, regression, domain adaptation, physics-informed learning
- **Frameworks:** PyTorch, LibTorch, TensorFlow, Scikit-Learn
- **Imaging:** ITK, VTK, ANTs, OpenCV; MRI/PET/CT/Ultrasound pipelines
- **Systems:** Linux, Docker, Git, Slurm; GPU profiling and deployment on NVIDIA platforms